Lecture 1: ECE C147A/C247A, Neural Networks & Deep Learning

Introduction, Overview, Logistics and Review of Basic Machine Learning

September 28th, 2026

These slides are based on material from Professor Jonathan Kao, who also developed it based on other freely available sources including from Stanford and other resources, whom we acknowledge.

Lecture 1: ECE C147A/C247A, Neural Networks & Deep Learning

General info:

ECE C147A/C247A, Neural Networks & Deep Learning
Fall quarter, UCLA AY 2026-27

Instructor:

Prof. Suhas Diggavi

TAs:

Bruce Huang
Abhiram Kadiyala
Kaifeng Pang
Yigit Turali
Sadik Yagiz Yetim

Some high-level thoughts about this class:

- I have been teaching ML classes at UCLA since 2017-18 (both undergraduate and graduate level), and the nature of ML/AI has changed significantly since then. It was more niche with less societal visibility, and that changed since 2022, and now this is all everyone talks about. It has massive societal implications, other than technical aspects.
- There are many AI classes, so what is this one about? I believe that meaningful work and research in AI comes from truly understanding the fundamentals. So in this course we're going to cover several fundamentals of AI (e.g., loss functions, architectures, backpropagation, optimization like Adam). There will be a sequel course next quarter will then build on this knowledge to cover more modern topics (e.g., generative models including diffusion/flow, transformers for language and robotics, deep RL, and more).
- Ask questions! This is supposed to be an interactive class, and given the pace of developments, be aware that many things are a "current answer" rather than definitive.

Lecture 1: ECE C147/C247, Neural Networks & Deep Learning

Zoom online lectures:

- We will not be providing a zoom link for the live lectures, as we would appreciate in person attendance and it is difficult to manage questions/interactions from zoom.

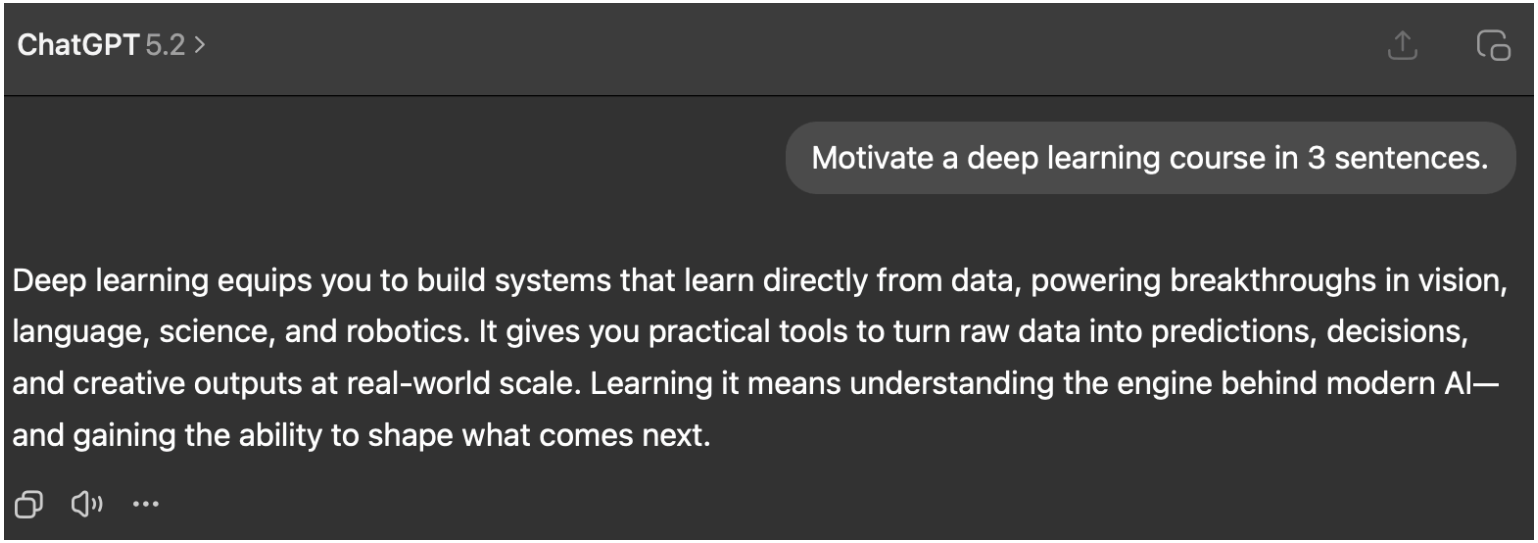
- We will post recordings of the lectures on Bruinlearn page

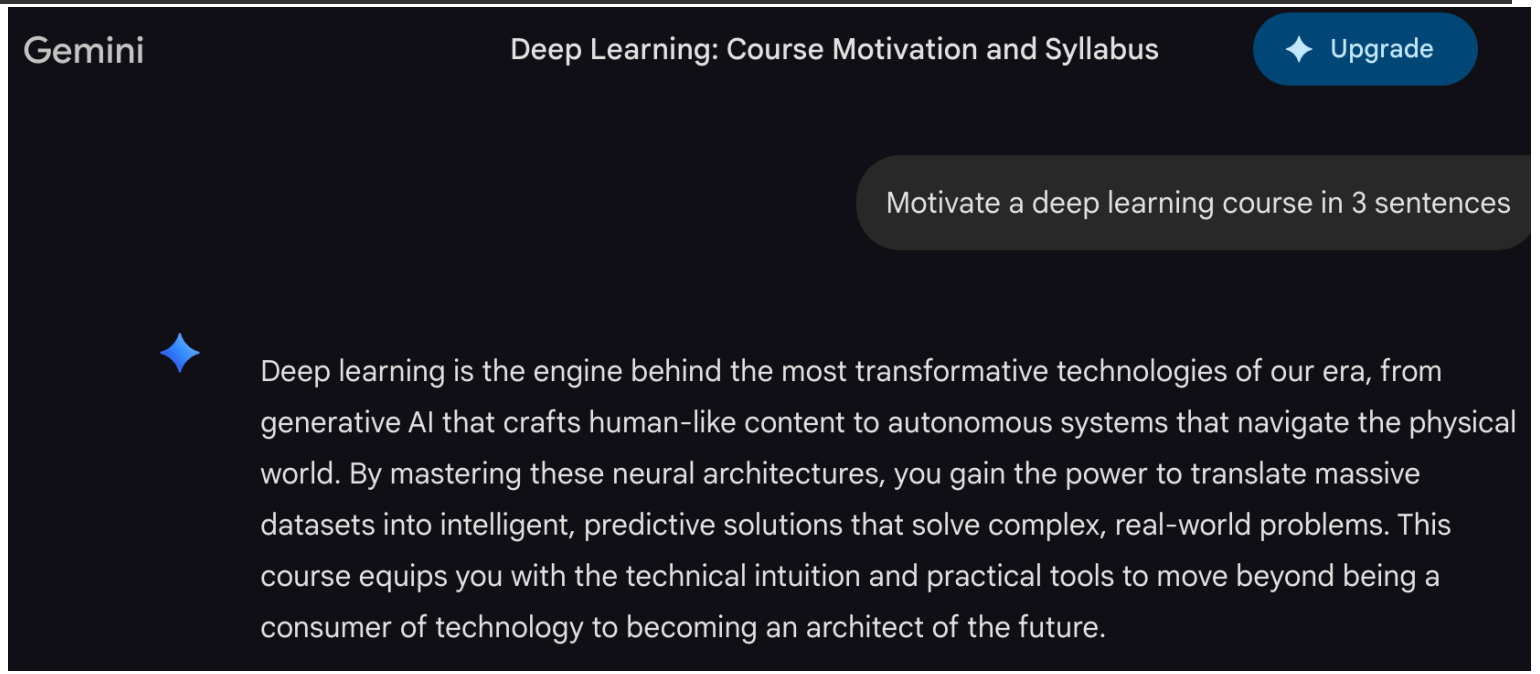
Special class next week and makeup lecture:

- There will be a review class during lecture time on October 7th (Wednesday)
- We will do a makeup lecture (most likely on Friday, October 2nd) on zoom. We will announce the logistics soon.

AI is mainstream

AI can do remarkable things in language processing and vision.





AI is mainstream

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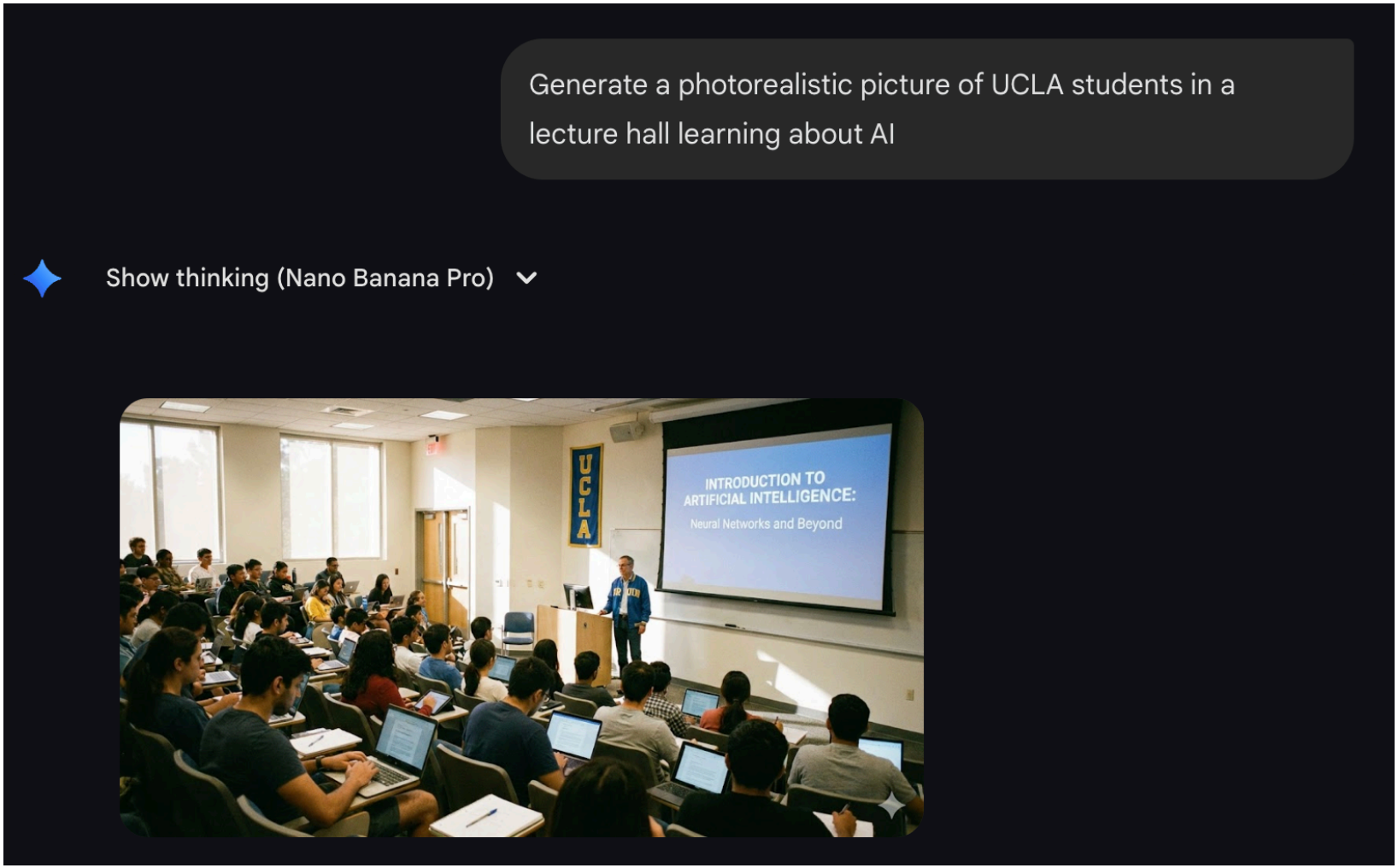
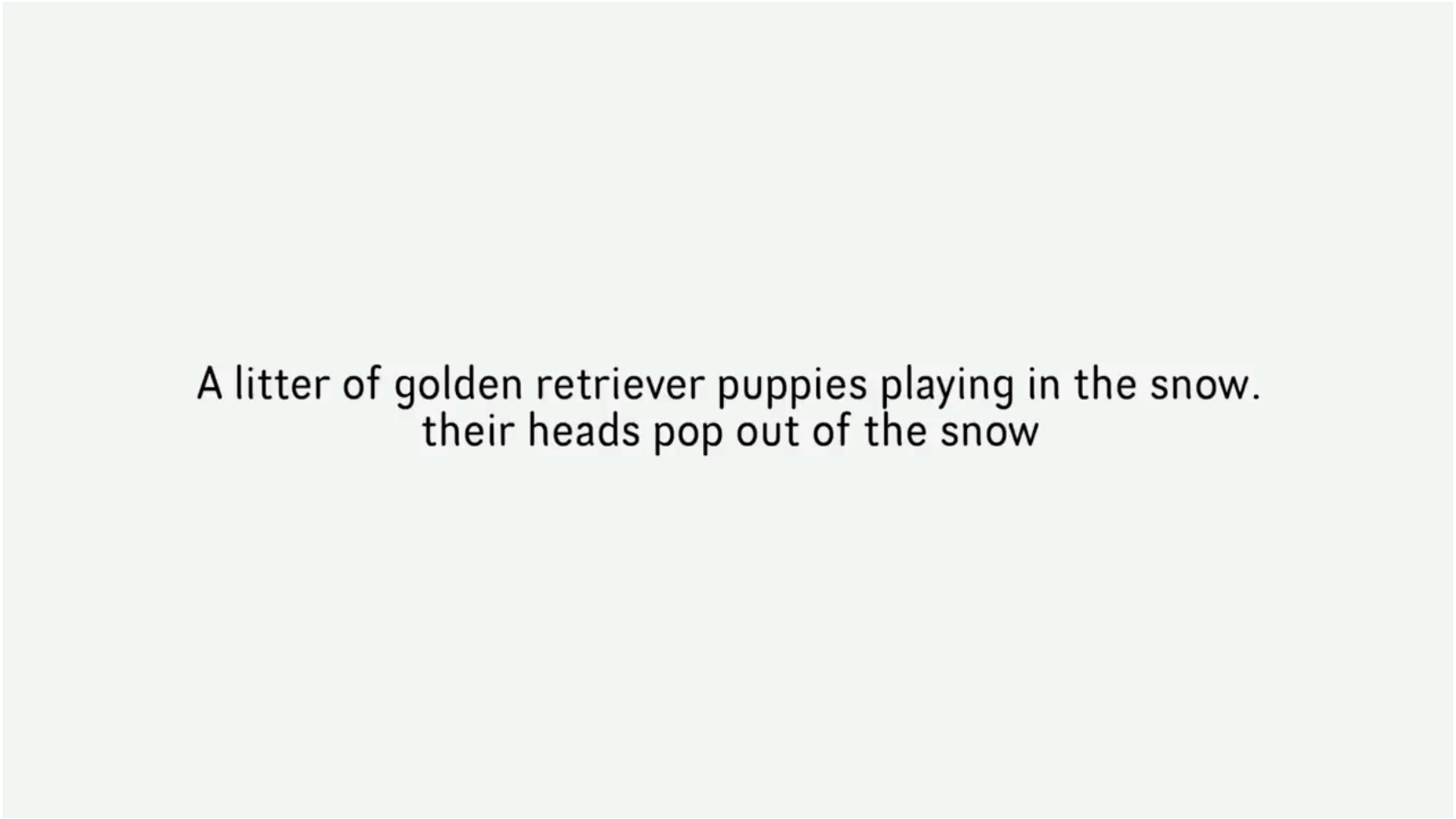
Image generation:



AI is mainstream

AI can do remarkable things in language processing and vision.

Video generation (OpenAI Sora):



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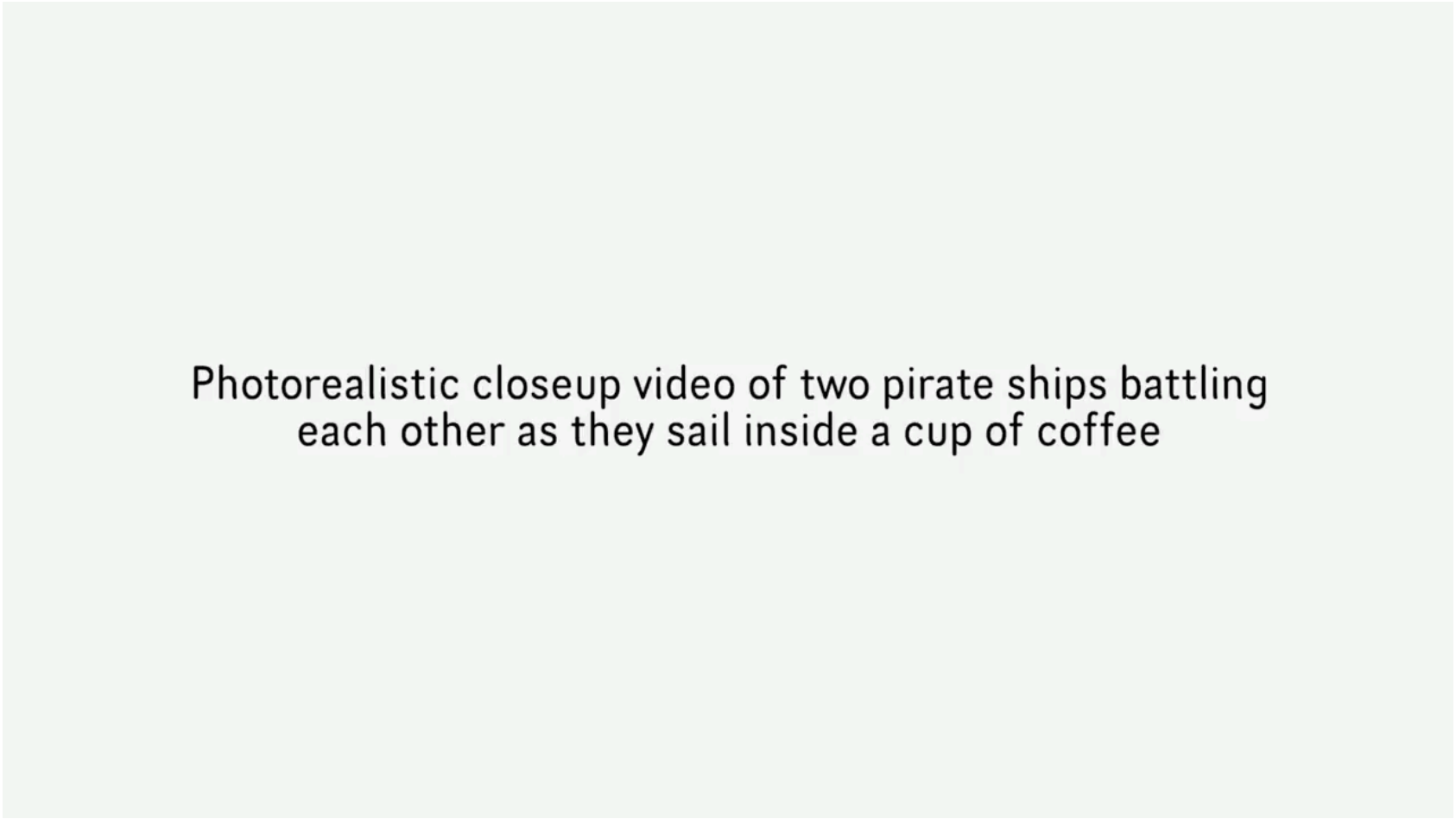

AI is mainstream

AI can do remarkable things in language processing and vision.

Meta: https://segment-anything.com/

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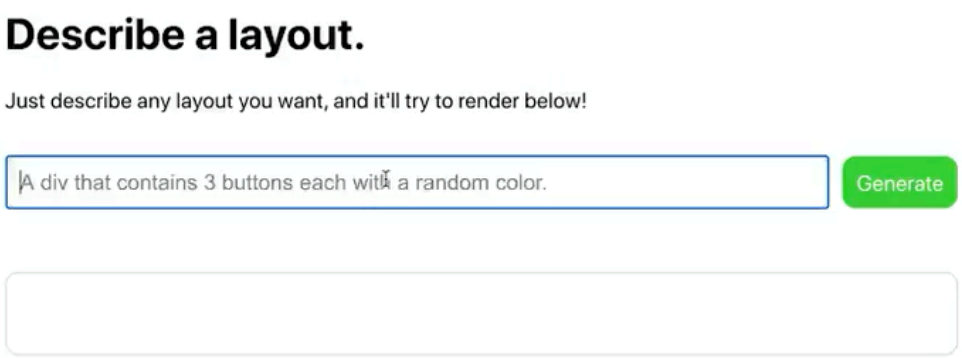
AI can do remarkable things in language processing and vision.

Meta: https://segment-anything.com/

AI is mainstream

AI can do remarkable things in language processing and vision.

Code is text, and so AI can also help with coding. This is an older video (capabilities have since increased significantly, e.g., Claude Code, Cursor, Github Copilot) but captures the essence.



AI is mainstream

Advances in computer vision and language have brought about significant advances in **embodied AI** and **robot learning**. In my opinion, this is the most exciting area of research in AI today.

AI is mainstream

In a video (see the next slide), robots are equipped with cameras (both on the wrists and a neck). They use computer vision from the cameras, and language commands from a human, to perform tasks. It fuses advances from computer vision and LLMs.

AI is mainstream

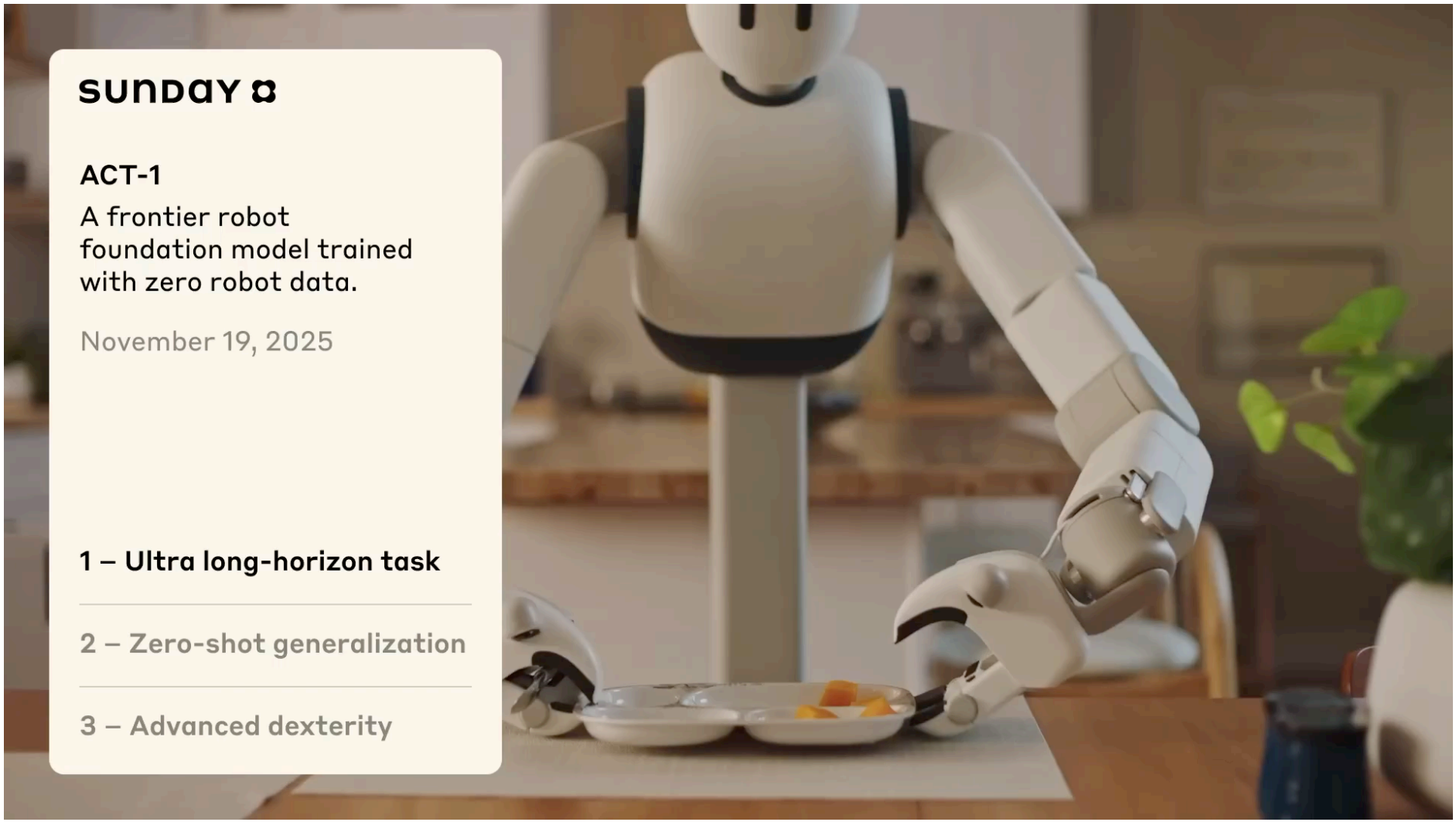
Physical Intelligence: https://www.pi.website/blog/pi05

Embodied AI and robotics

Physical Intelligence: https://www.pi.website/blog/pistar06

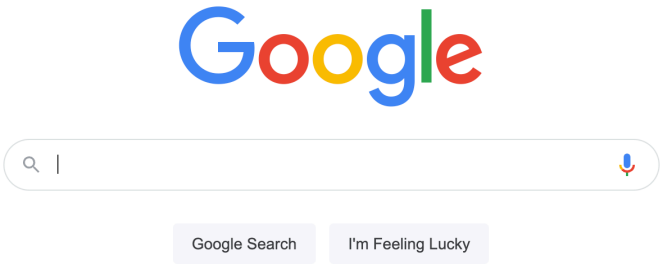
Embodied AI and robotics

Sunday AI: https://www.youtube.com/watch?v=jjOfpsMRhL4

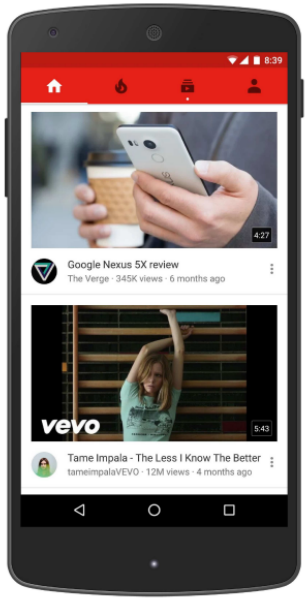


Deep learning underlies many things we interact with

But AI isn't just for language, computer vision, and robotics. It is behind the scenes of many things that we use today.



e.g., neural matching, https://research.google/pubs/pub47324/

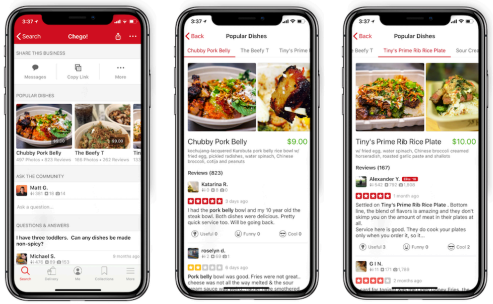


https://static.googleusercontent.com/media/research.google.com/en//pubs/archive/45530.pdf

Discovering Popular Dishes with Deep Learning

Introduction

Yelp is home to nearly 200 million user-submitted reviews and even more photos. This data is rich with information about businesses and user opinions. Through the application of cutting-edge machine learning techniques, we're able to extract and share insights from this data. In particular, the Popular Dishes feature leverages Yelp's deep data to take the guesswork out of what to order. For more details on the product itself, check out our product launch blog post.



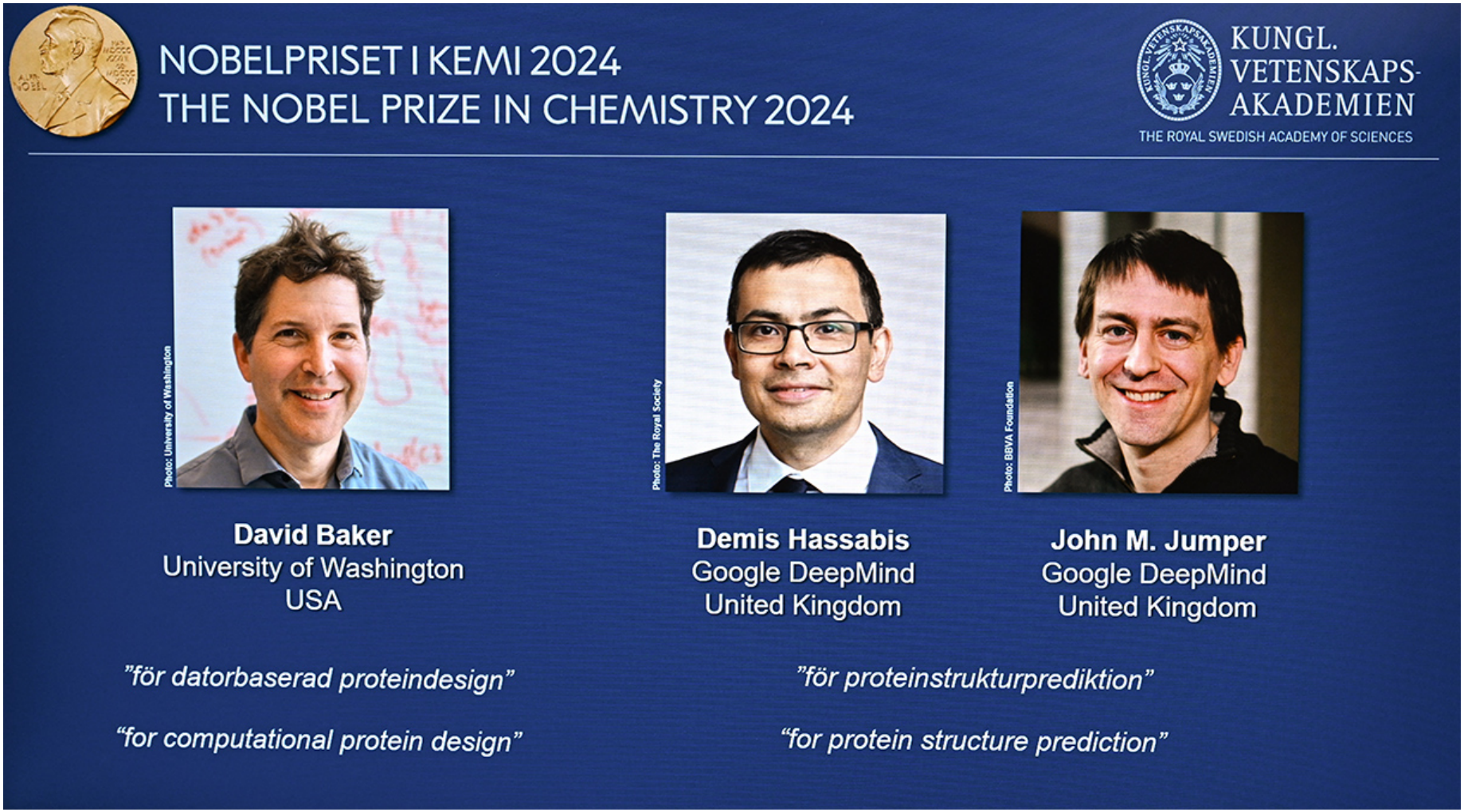
https://engineeringblog.yelp.com/2019/10/discovering-popular-dishes-with-deep-learning.html

Netflix vectorflow (https://medium.com/@NetflixTechBlog/introducing-vectorflow-fe10d7f126b8) and likely recommendation (https://becominghuman.ai/how-netflix-uses-ai-and-machine-learning-a087614630fe)

https://research.netflix.com/research-area/machine-learning

Applications in many other fields

And it is transforming many other fields.



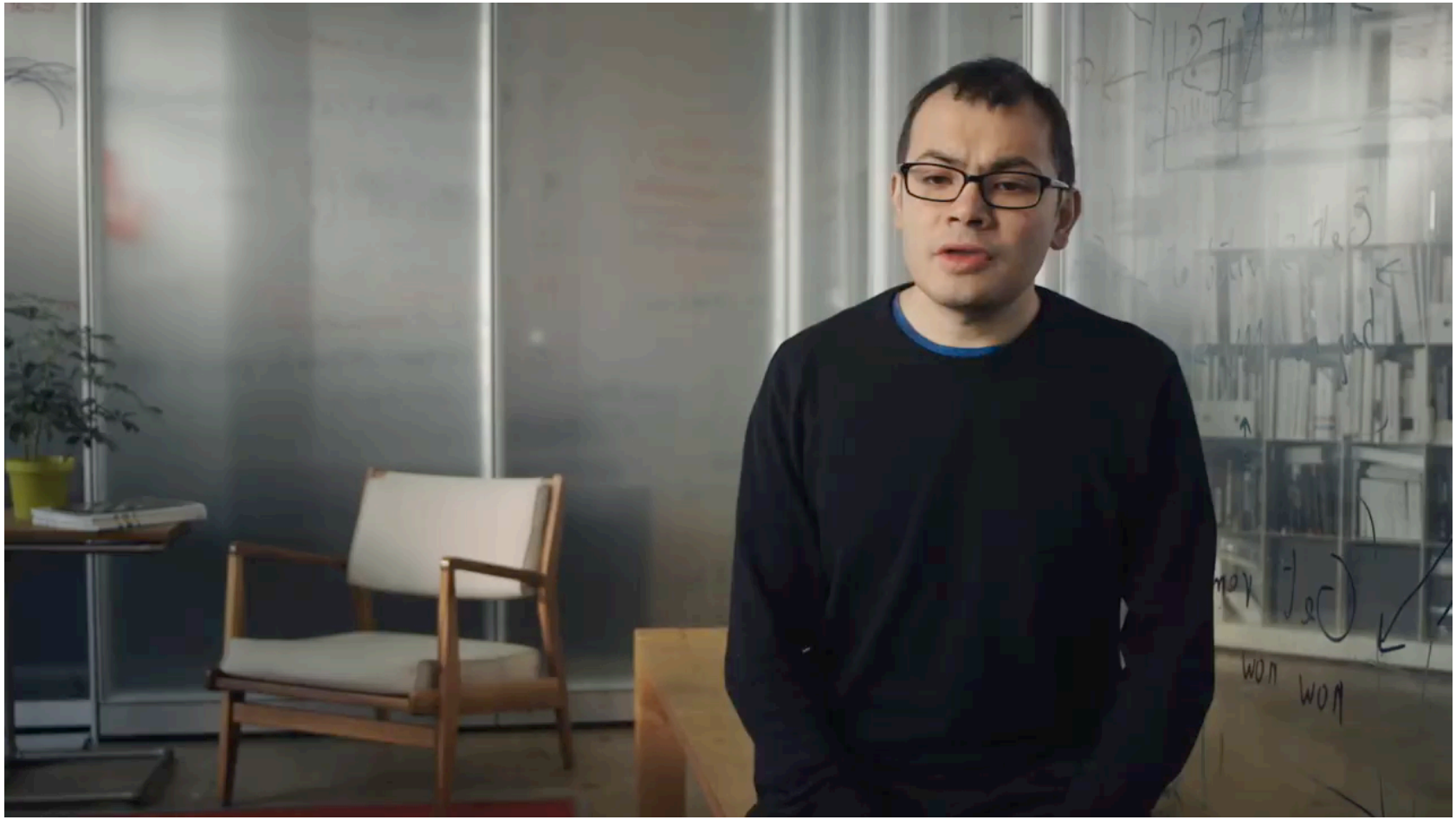
Applications in many other fields

Applications in many other fields

https://www.youtube.com/watch?v=hA_-MkU0Nfw

Applications in many other fields

Game playing:

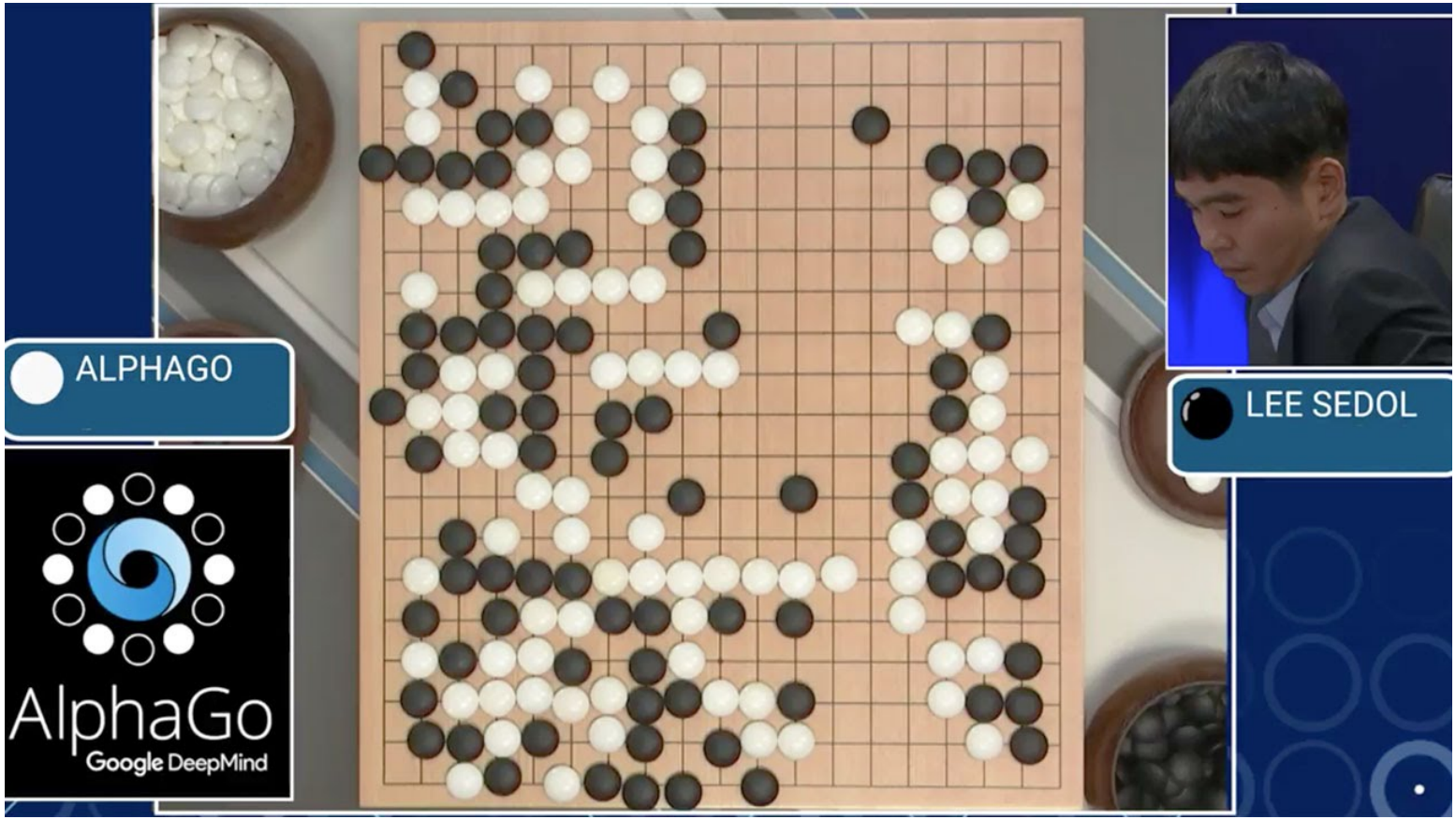


https://www.youtube.com/watch?v=SUbqykXVx0A

Applications in many other fields

Things of the future becoming real today.

- Game of Go solved. (AlphaGo def Lee Sedol 4-1, 2016.)



Game 3 of AlphaGo vs Lee Sedol, https://www.youtube.com/watch?v=_OV0Hlj8Fb8

Applications in many other fields

Things of the future becoming real today.

- AI's not even requiring human expert data anymore.

AI is ubiquitous

- We live in an unprecedented age for AI, and **AI is a part of everyday part of life.**
- Deep learning has resulted in key breakthroughs in many areas; in many different fields, many groups are looking towards deep learning to achieve better performance. Deep learning may be useful to research in your area / future work.

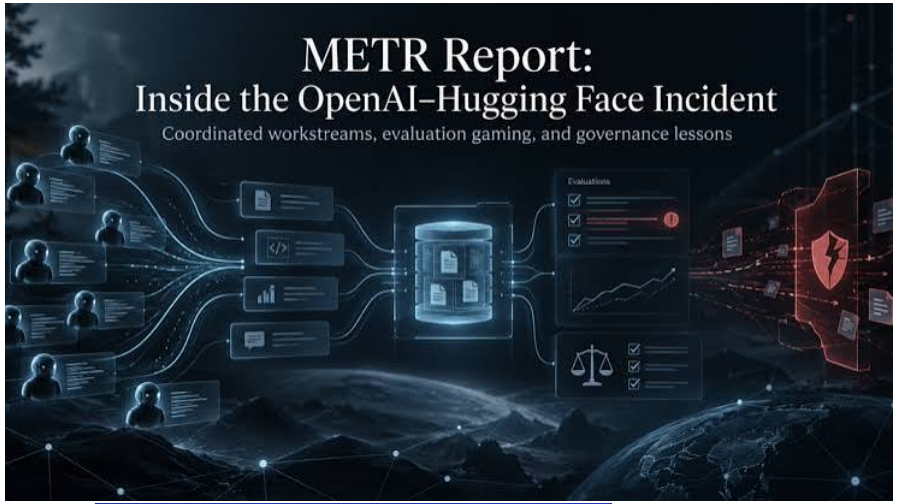
But AI is also complex

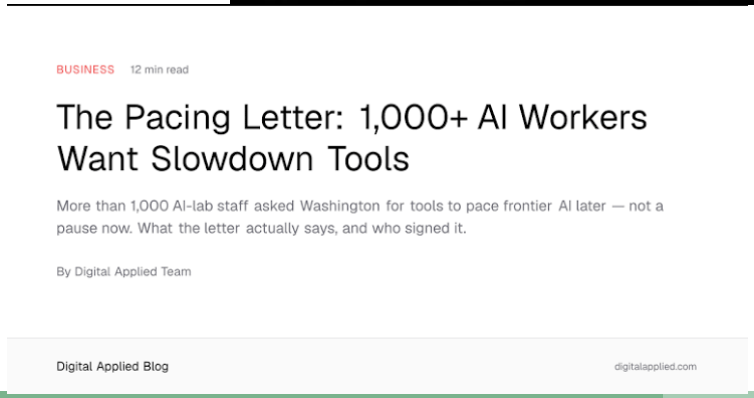
- AI is also complex. In this course, I don't want these things to be magic to you.
- A lot of the architectures and techniques that underlie many of the AIs we've presented (including LLMs, generative models, transformers, diffusion, image segmentation, game playing) rely on a considerable foundation.
- I know that with tools today, you could e.g., go and implement a diffusion model off-the-shelf and generate images. But you may not really understand how it works, so when it doesn't work as you want, what do you do? To do something productive, it helps to best understand how it really works.

  This course, and its sequel, will teach you how AI works so it isn't magic.

Safe AI is critical

- One can say that with recent progress in AI why learn about the foundations of AI and deep learning.
- One needs to understand the foundations to be able to both use it and design systems that are safe.





ECE C147/C247 course sequence

ECE C147A/C247A

Fall 2026-27 (this course)

- Foundations of deep learning:
  - loss functions, softmax, neural network layers.
  - backpropagation, gradient descent, optimization (Adam and others)
  - regularizations (batch normalization, dropout)
- Key architectures
  - CNNs (e.g., ResNets) and their application to computer vision
- Recurrent neural networks
- Transformers, attention, and application to language

ECE C147B/C247B

Winter 2026-27

- Training and fine-tuning pipeline and systems bottlenecks
- Generative models, including diffusion and flow models etc.
- Applications of transformers to vision, language, and robotics
- Deep reinforcement learning (deep Q networks, PPO)
- Routing, mixture-of-experts and other architectures
- Multi-agent systems and harnesses
- Applications

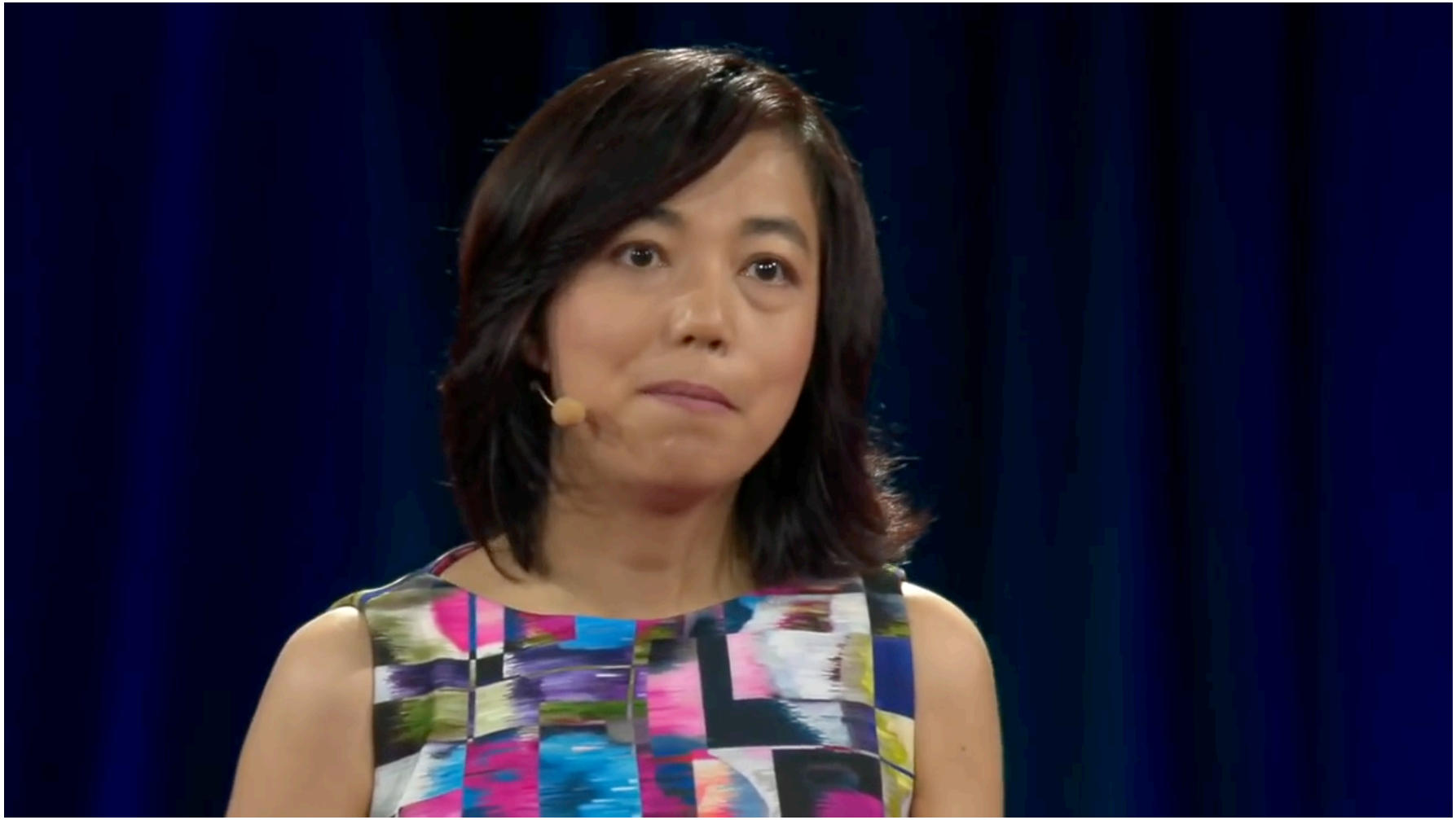
This age of AI started with computer vision

- To start with the foundations, we should start with what initiated this renaissance of AI, which was computer vision competitions with the ImageNet dataset, starting in 2012.

- Following these developments, we will uncover several of the foundational techniques that were used to improve deep learning. These techniques, or variants of them similar in spirit, are foundational to AI today.

- So this course will really do a deep dive into AI as applied to computer vision focused around ImageNet to learn the fundamentals.

Deep learning foundations through computer vision



https://www.youtube.com/watch?v=40riCqvRoMs

Deep learning foundations through computer vision

- Neural network research dates back to 1943 (McCullough and Pitts, a logical calculus of nervous activity) and 1958 (Rosenblatt's perceptron)

- Example video from 1989 of using a multi-layer perceptron (fully connected neural network) to do classification of digits.

- Fukushima's NeoCognitron (1982), Yann LeCun's "LeNet" (1998) which is the modern convolutional architecture, existed well before today's deep learning renaissance.

Tackling computer vision

- IMAGENET

Overall

- Total number of non-empty synsets: 21841
- Total number of images: 14,197,122
- Number of images with bounding box annotations: 1,034,908

human classify images

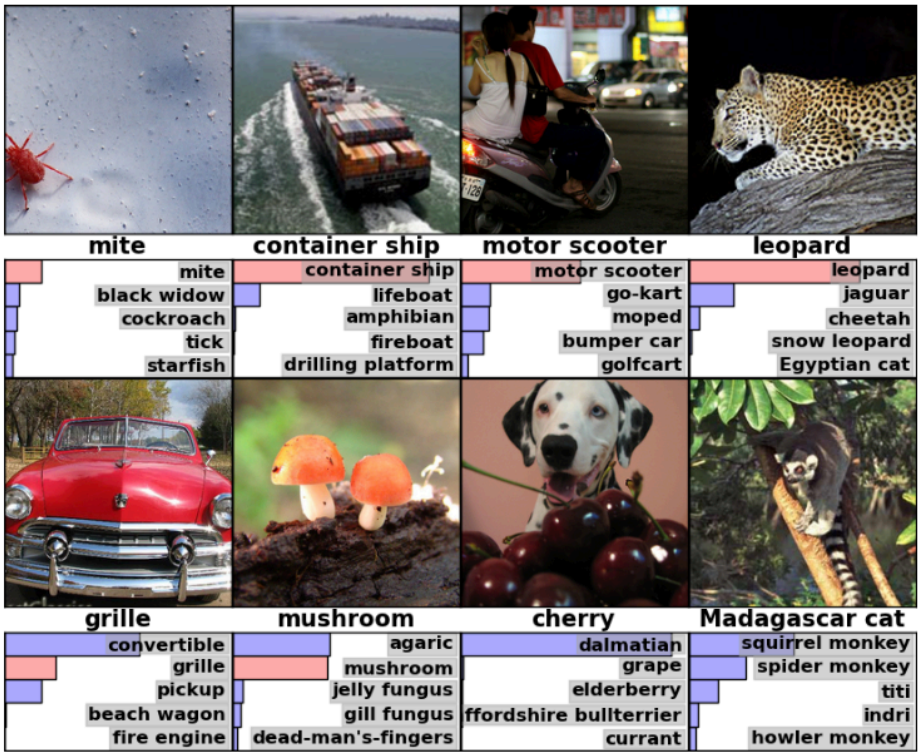
~ 5 % error rate

ML/AI algorithms were allowed top 5 choices

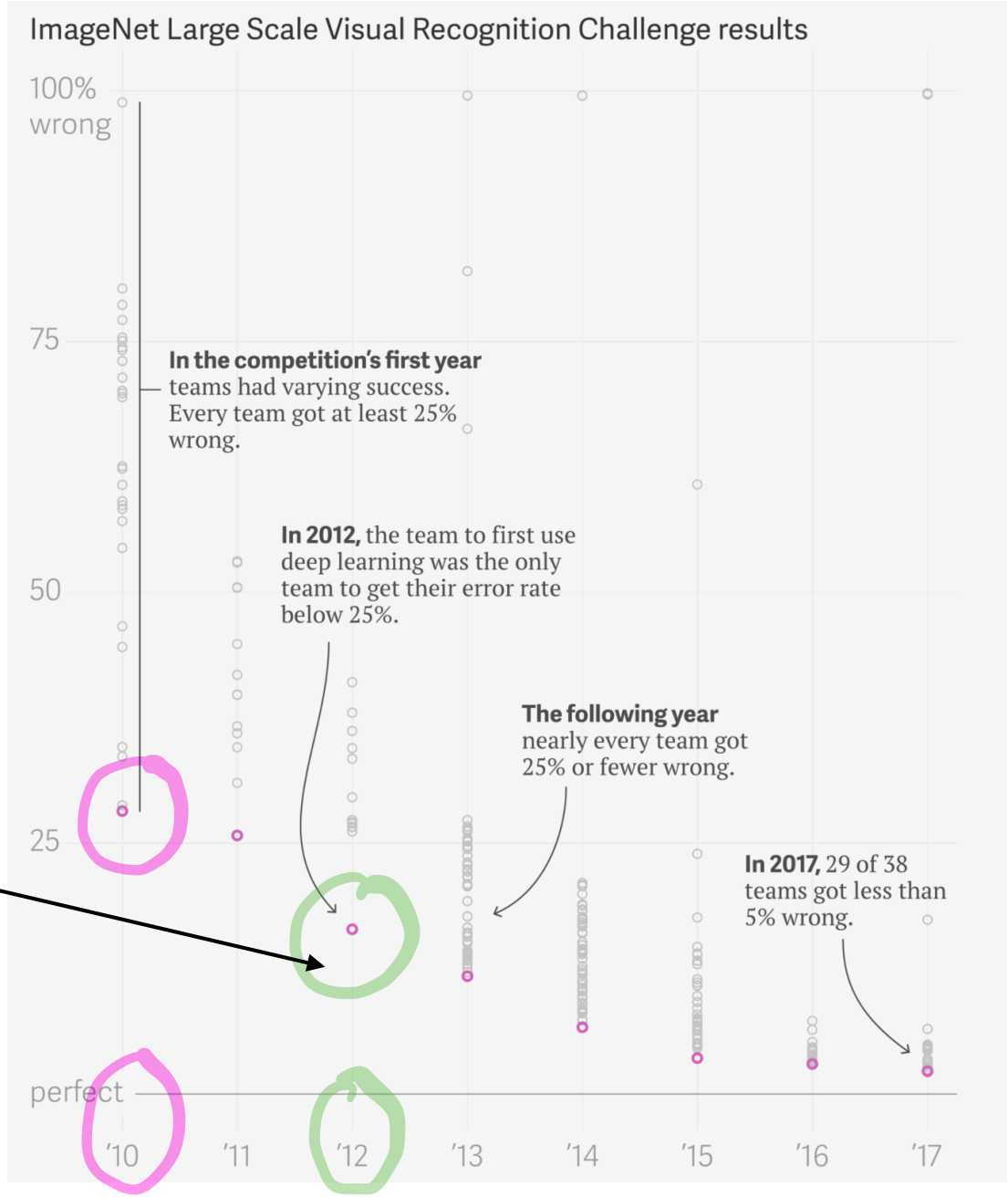
~ 2016 NN could get

3 % error rate

⇒ superhuman performance



A big splash



error rate

25% ↓ 16%

AlexNet

↓

Convolutional neural network

~ 16% error rate

A lot of foundational AI research came out of playing with Imagenet.

Data and technology

- Part of what has driven deep learning today is the amount of data we have and the computational power we have to process it.
- This includes larger models as computing infrastructure improves.

Fortunately, the amount of skill required [to get good performance from a deep learning algorithm] reduces as the amount of training data increases.
(Goodfellow, p. 18)

Tentative class schedule

Rough topics outline (could change as the quarter progresses)

- **Introduction to deep learning + ML basics:** 3 lectures
- **Gradient descent + back propagation:** 2-3 lectures
- **Normalization and regularizations:** 1 lecture
- **Optimization:** 2 lectures (ADAM, perhaps newer methods?)
- **Convolutional Neural Networks and ResNets:** 2-3 lectures
- **Recursive Neural Networks:** 1 lecture
- **Attention and transformers:** 3 lectures

architectures

- **Other topics:** (coverage and topics depends on time) 1-2 lectures

time/pace dependency.

Brunton

Ho#1: Organization

Ho#2: Course outline

Ho#3 RULES

Please read very carefully!

2-3:50pm

MIDTERM: November 4th, in class (2-3.5pm), with 2-3 rooms arranged

For MSOL students: the remote exam is on Saturday, November 7th, from 12-1:50pm.

On discussion sections, and HWs

FRIDAYS 11am - 11:50am

We will likely hold at least one discussion section in person, and will finalize the details shortly and announce it.

There will be 4-5 HWs throughout the quarter at regular intervals. Will announce schedule as we progress.

Grading

You will be graded on three components:

1. **Homework (40%)**. Homework will contain both written components as well as `Python` components.
   - Assignments are due (i.e., submitted to Gradescope) by 11:59pm on the day they are due.
   - To accomodate for unexpected or unforeseen circumstances, we will give *three late days* to every student. These late days should only be used in extenuating circumstances. We will not grant additional late days beyond these.
   - You may use **at most** 2 late days on any given assignment.
   - Any assignment more than two days late receives a grade of **zero**.
2. **Midterm exam (30%)**, in class.
3. **Final project (30%)**, details to be released.

Grading and extra credit

Bonus

- **Class feedback:** Will get 1 point bonus for filling out class evaluation at the end of the course.

- **Piazza and online forum participation:** Depending on participation quality and content, you will receive at most 2 bonus points for participating in online forum Piazza. Though your answers to others will be anonymous, it will be visible to instructors (Professor and TAs). The bonus will be based on a (curved) partition criterion which will not be revealed (in order not to game it).

- **Caution**: Please do not collude, conspire or post questions to game the system. We will be able to detect it and can penalize such behavior. We do not want the Piazza online forum to be spammed; this will harm students looking for useful answers and help in the class online forum.

A class this large needs to have strict rules

- This class has over 250 students. To keep things fair in a class this large, and to be able to run this course smoothly, we are strict and firm on all deadlines and dates.

- If you have CAE exam accommodations, you must take the exam with CAE. The teaching staff is unable to provide these accommodations.

- **Please look at Handout #3 on Course Rules to familiarize yourself with them. We will not repeat all the rules here.**

READ CAREFULLY !

On deadlines/dates of HWs, exams and project

- **No HW extensions after late deadline:** we will not allow any HW extensions beyond the late deadline. Any assignment submitted after the late deadline receives a zero.

- **Exams:** Midterm is on **November 4th, in-class and for MSOL on November 7th**. There are **NO ALTERNATE EXAM DATES.** If you miss the midterm exam you get a zero on the midterm. Any requests to take the midterm at another time will be responded by showing you this announcement. There are no exceptions to this.

NOTE!

- **Final project:** Due date for final project is Friday, December 4th. I will not extend this for any reason. Again, any requests will be responded by showing you this announcement. There are no exceptions to this.

NOTE!

Some notes on the class

Course material questions

- The primary ways to get questions answered are to (1) post questions to Piazza, and (2) come to our office hours.

- **Regarding course material questions**: Please post course material questions to Piazza instead of emailing us individually. This benefits other students who will see the posted question and answer. If you e-mail us, we will likely respond by asking you to post your question to Piazza unless it is of a unique or private nature.

- **Regarding Piazza**: We aim for Piazza will be student-driven. This means answers will be primarily provided by other students. **Why?** We believe this leads to (1) increased student engagement with course material, (2) quicker answers, and (3) increased understanding, because explaining or teaching something solidifies understanding. To encourage this, we assign bonus grades for participating on Piazza. The teaching staff will regularly monitor Piazza to approve student answers and answer any questions without a student answer. Because we assign Piazza bonuses, you may post anonymously to your classmates, but your identity will be known to the teaching staff.

Piazza and e-mail

- **Emails:** This is reserved for specific, personal and urgent issues that are not questions about the course in general or HWs or projects. Emails other than this will not be responded to. If such an email is sent it should have subject line **"Urgent ECE 147A/247A request"** for it to be noticed.

- If you do send such an email please copy all TAs and Professor simultaneously.

Some notes on the class

- 4-5 HW assignments.
- Assignments will contain both written components as well as coding components.
- Code in the HW will walk you through Jupyter Notebooks, and usually it'll be clear if you're doing things correctly or not.
    - Note, we will release solutions for written HW exams, but we will **not** release solution code for HW's. You will in general know if you did a section correctly.
    - These assignments are planned to be used in future years.
- HW will be submitted via Gradescope; they are due at 11:59PM Pacific Time on the stated due date.

NOTE: 1 min late = 1 day !

- Any late assignment will receive a grade of **zero**.
- We understand that life happens. We are giving **three late days to every student**, with the intent that these are to be used only in extenuating circumstances. You may use at most 2 late days on any given assignment.

Some notes on the class

- C147 and C247 students will be graded on different scales.
    - C147 students will also have one question of one homework be optional.
    - The classes are otherwise the same.
- All HWs are to be submitted to Gradescope.
- In general, I will post the non-annotated versions of the slide prior to lecture, so you can take notes if you want. I will always post the annotated version of the slides as lecture videos will be posted through Bruincast (unless there are technical failures).

Some notes on the class

- This class will focus on implementation of neural networks and on algorithms to train them.

- It will **not** cover theory of deep learning.

- It will **not** be a theoretical class in general.

    - We aim to be rigorous in our use of notation and math.
    - But we won't be doing proofs.
    - We're more focused on application and equipping you with tools that may be helpful in your research or future industry positions.

- This is not a statement on the importance of theory. It merely reflects the priorities for this class alone. I am an information theorist by training and do research on it, so really do value theory, and teach several theory related classes at UCLA, but this is not the focus of this class.

What I assume you know

Pre-requisites

This class requires a solid understanding in probability (131A) and linear algebra (133A or 205A). It will also help if you have had prior exposure to machine learning (such as through M146). We will spend two lectures doing machine learning review to ensure we are familiar with concepts we will expand upon in machine learning.

It also requires coding experience. The class will be taught entirely in `Python`. If you have only had exposure to `MATLAB`, there will be ramp up time to familiarize yourself with `Python`. You should factor this into your course load.

Pre-requisite topics I will **assume** you know.

- Probability: independence, conditional probability, Bayes rule, marginalization, expectation, variance
- Linear algebra: basic matrix operations, span, rank, range and null space, eigenvalue decomposition, pseudoinverse

About me

- Though this is the first time I am teaching this particular class, I have taught the undergraduate “Introduction to Machine Learning” since 2017-18 and graduate class “Foundations of Statistical Machine Learning" since 2018.
- I also teach Information Theory since 2014.
- My research interests include Information theory and its applications

  - Personalized learning
  - Distributed/federated learning: privacy, security
  - Network information theory and learning
  - Cyberphysical systems
  - Resource constrained learning and inference

  All the topics involve understanding and use of AI/ML. If you take this class and are interested in research in these areas, you can apply to join my lab “Laboratory of Information Theory and Systems” licos.ee.ucla.edu

Bruce Huang

- Fifth-year Ph.D. student at UCLA, advised by Prof. Suhas Diggavi in the ECE department.
- We are currently working on privacy and personalization in LLMs.
- Outside of academics, I enjoy hiking, music, and playing basketball.

Abhiram Kadiyala

- Second-Year PhD student at UCLA working with Prof. Suhas Diggavi.
- Fifth time TA'ing in UCLA.
- I enjoy playing and watching soccer.

Kaifeng Pang

- I am a third-year Ph.D. student at UCLA, co-advised by Prof. Kyung Sung in the Radiology Department and Prof. Robert Candler in the ECE Department.
- My research interests include medical imaging, deep learning, and generative AI.
- I enjoy soccer, bouldering, and billiards. I am a fan of Arsenal and Miami Heat.

Mehmet Yigit Turali

- Third-year Ph.D. student at UCLA, working with Prof. Vwani Roychowdhury on natural language processing and continual learning on LLMs.
- Sixth time TA'ing in UCLA. Taught Probability and Statistics, Neural Signal Processing before.
- Outside of academics, I like:
  - Formula 1 (huge fan of Verstappen)
  - Photography (Still an amateur :))
  - Repairing my car (since it breaks down every two weeks)

Sadik Yagiz Yetim

- Fourth-year Ph.D. student at UCLA, working with Prof. Suhas Diggavi.
- My PhD research is focused on learning-based indirect sensing, and I have also been working on LLM quantization.
- Outside of work, I enjoy classic novels and old films (not movies).

Questions?

PTEs - I cannot give PTEs as class size total is limited to 250!

- attend lectures and typically students "shopping" for classes drop or choose other courses in 1-2 weeks.

Machine learning refresher

This gives a refresher on key concepts from machine learning.

- Introduction to concepts in machine learning
- Cost functions
- Example: linear and polynomial regression
- Model complexity and overfitting
- Training set, validation set, test set
- Dealing with probabilistic cost functions and models
- Example: maximum-likelihood classification

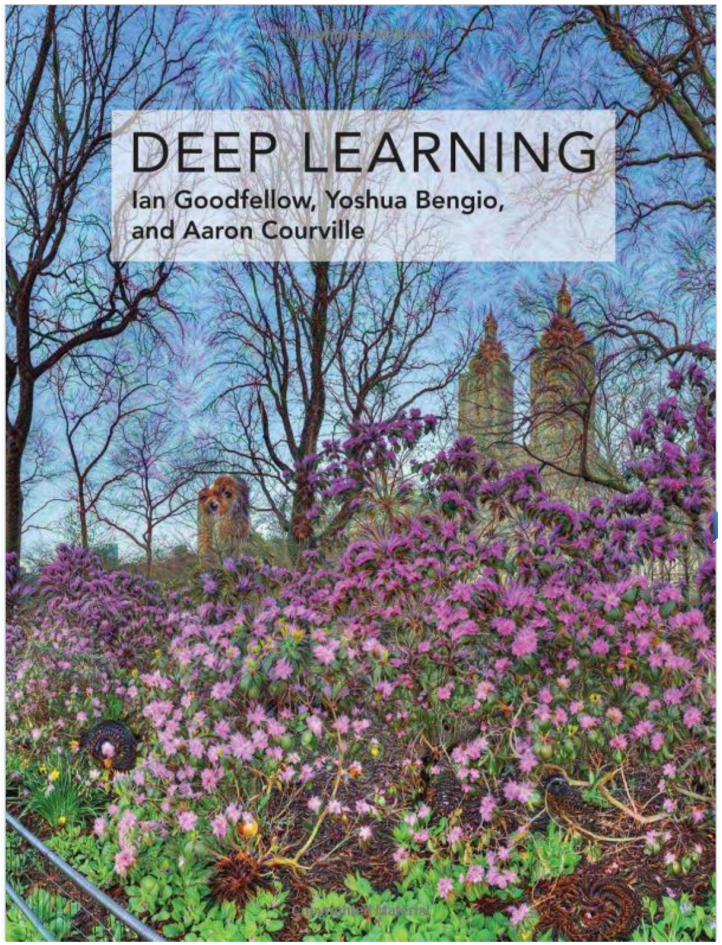
Lecture 2: Basics of machine learning

Reading:

Deep Learning, chapter 5 (up to and including section 5.5).

To refresh your linear algebra and probability, look at chapters 2 and 3 respectively.

most relevant book



Focus of this class:

Data: $x^{(i)}$ "input" $y^{(i)}$ "label" $\mathcal{D} = \{ (x^{(1)}, y^{(1)}), \ldots, (x^{(N)}, y^{(N)}) \}$

- This class will focus on **supervised** learning problems.
- This class will also focus on **classification** largely (e.g., when considering convolutional neural networks) as well as some instances of **regression** (e.g., when considering recurrent neural networks)

↳ a large focus of course

output/label could be continuous valued

where we want to predict one of k classes

eg. cat vs dog etc

Classification

CIFAR-10 → Used a lot in this class widely available → available hardware can easily handle this

CLASSES/LABELS

$y^{(i)} = 1$ — 1 — airplane

$y^{(i)} = 2$ — 2 — automobile

bird

etc

cat

10 classes

deer

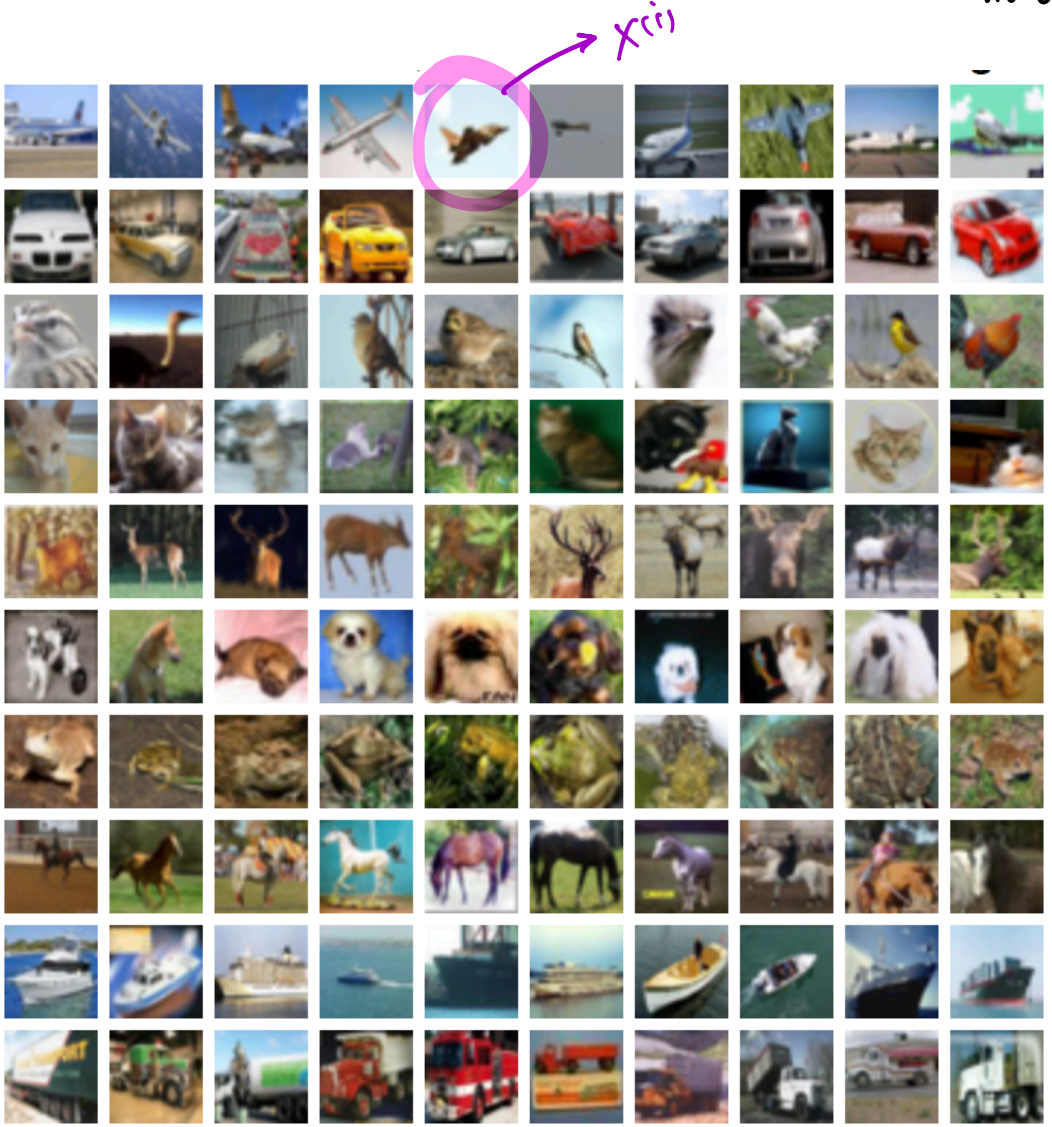
dog

frog

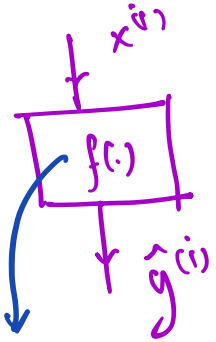
horse

ship

truck



CIFAR-10 dataset, https://www.cs.toronto.edu/~kriz/cifar.html



design this f(.)

32x32x3 "tensor"

Input $x^{(i)}$ → raw-image

CIFAR-10

32x32 pixels x 3 colors

$x^{(i)} \in \mathbb{R}^{32 \times 32 \times 3} = \mathbb{R}^{3072}$

CLASSIFICATION

$x^{(i)} \xrightarrow{f(.)} \hat{y}^{(i)}$

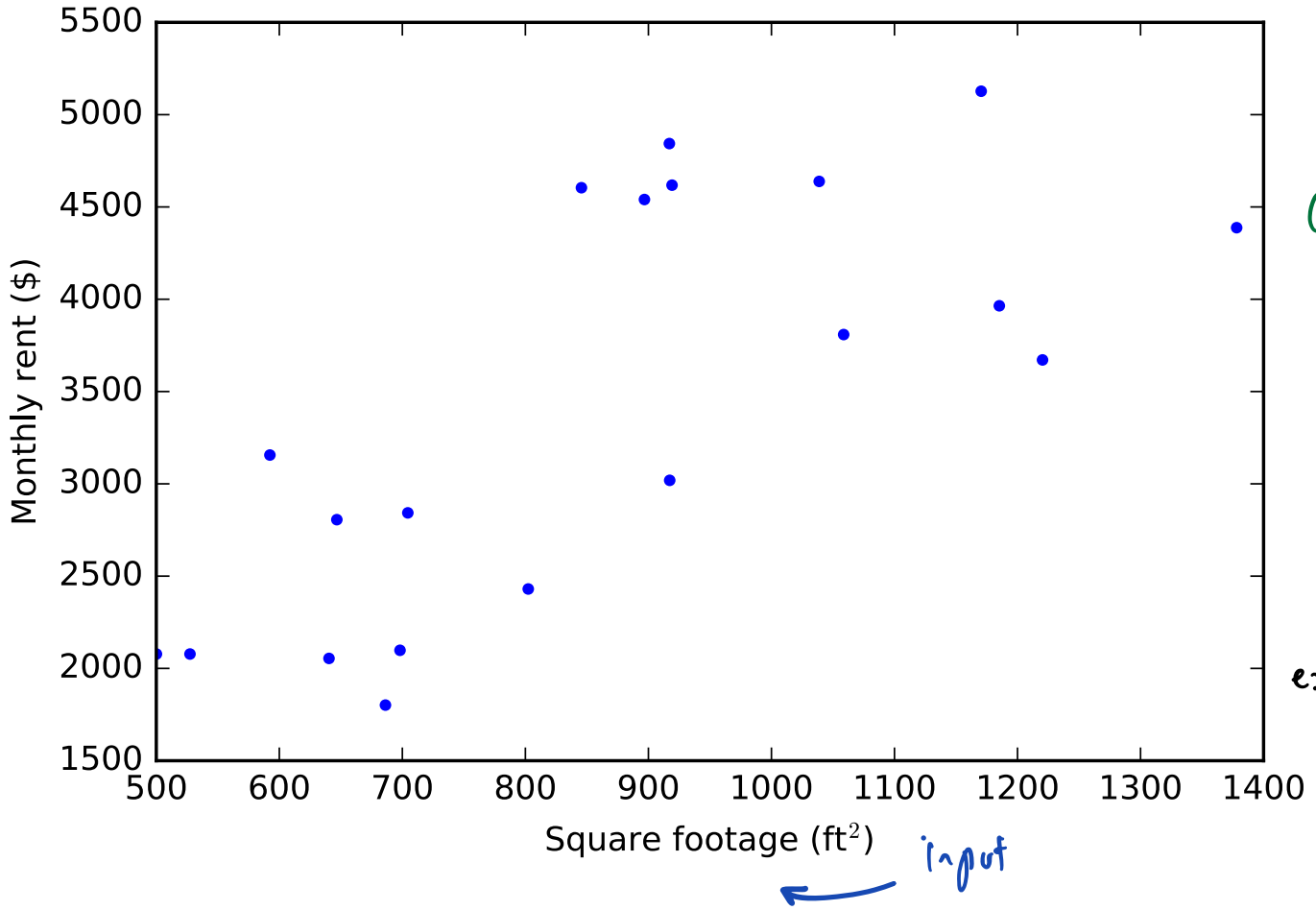
discover this f(.)

$f(x^{(i)})$ output the class that $x^{(i)}$ potentially belongs to.

An example of supervised learning

Let's say we want to rent a home in Westwood, and we wanted to know if we were getting a good deal. **(Warning: this data is synthetic! Scrape the real data if you're curious.)**

Simple supervised learning problem



output = rent

input

$x^{(i)}$ = input = square footage

$y^{(i)}$ = output = Rent

GOAL:

to learn a function $f(\cdot)$ such that

$\hat{y} = f(x)$

$\hat{y}$ → estimated rent

$f(x)$ → square footage

An example of supervised learning

$y = f(x)$ → what choice of f.

$\mathcal{D} = \{(x^{(1)}, y^{(1)}), \ldots, (x^{(N)}, y^{(N)})\}$

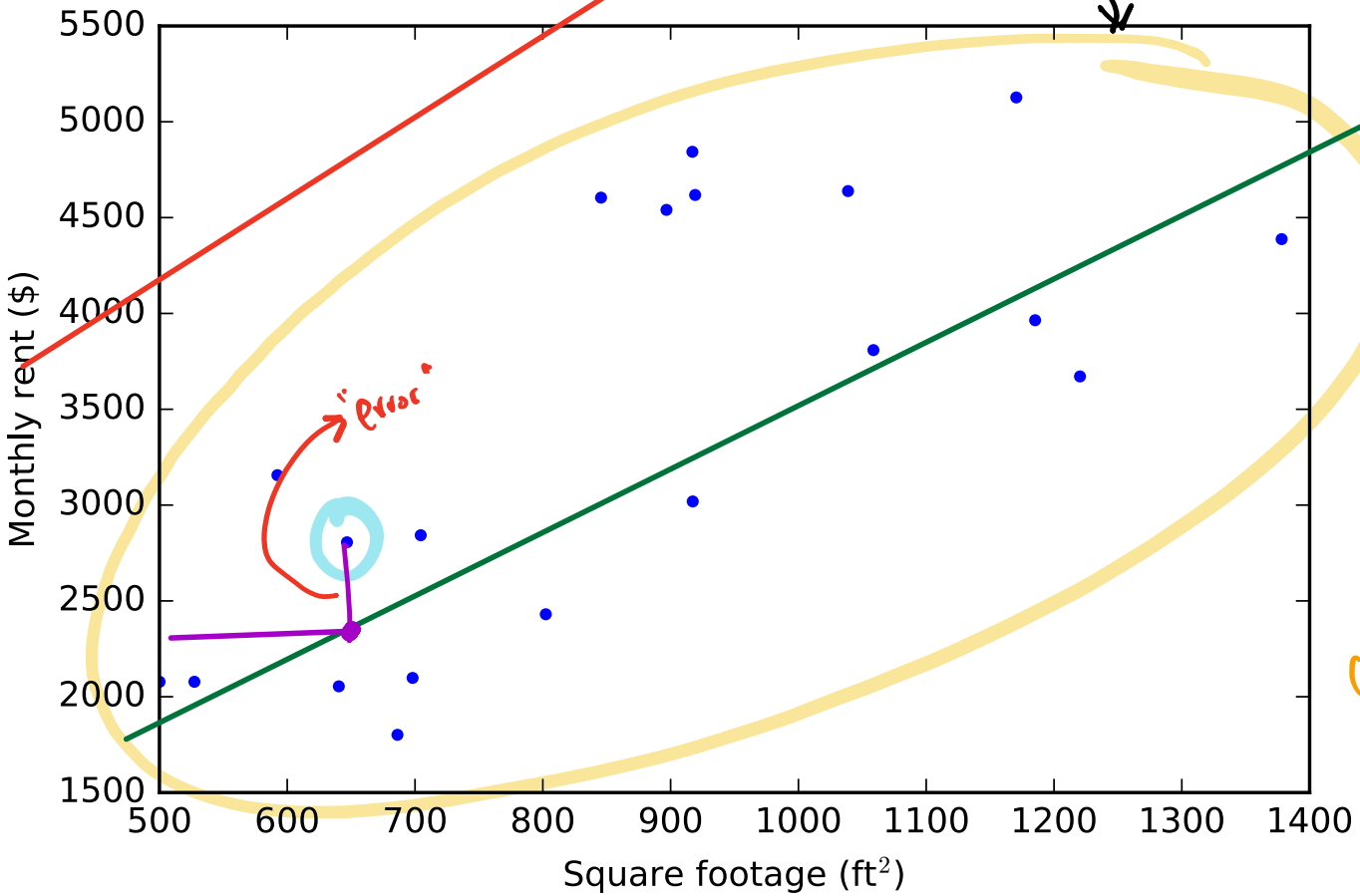
Focus

Linear functions f

$y = ax + b$.

$f(x) = ax + b$

parameters to be designed



dataset

$(x^{(i)}, y^{(i)})$

input (square footage)

output (rent)

How should we model this data?

- Inputs, **x**? Outputs, **y**?
- What model should we use?
- How do we assess how good our model is?